# Food Waste and Lost Sales in One Store, Q3 2026

Analysis and Recommendations for Ordering Optimization

Prepared by: Buruhani Mutebi | Period: 1 July to 30 September 2026

## 1. Summary

This analysis covers daily food operations from 1 July to 30 September 2026. The dataset includes daily stock movements, sales, waste (mark out) and stock-out events across all food categories. The focus is on identifying waste drivers, stock-out patterns, and ordering rule performance to reduce cost and lost sales.

Mark out is valued at cost price; no currency is stated in the data.

- Waste (mark out) was 21,435.83 at cost: 3,126 units over 92 days, about 233 per day. Sandwiches and pastry are 53% of it. Salads and muffins have the highest waste rates (26.7% and 22.7% of stock handled).
- Stock-outs: 671 of 3,894 item-days (17.2%) ended with nothing left and no mark out. 19 of these were days with no stock at all (certain lost sales); 652 are closing-zero days (possible lost sales).
- Waste is highest early in the week (Tuesday 394.04 per day, Monday 346.29) and lowest at the weekend (Friday 145.69). Stock-outs are the reverse: Friday 24.9% of item-days, Monday 7.5%.
- The stated 10% / 30% ordering rule cannot be confirmed from the log alone. Only 22.4% of 2,638 orders match it exactly. Total units are close (-0.8%), but the day-by-day pattern differs, and the log has no noon sales, which are needed to rebuild the orders.
- A replay of stock under alternative rules shows a trade-off, not a clear winner. Ordering the average of the same weekday over the last 4 weeks cuts waste by about 16% but leaves 167 item-days where recorded sales could not be met. No simple rule tested cut waste without missing recorded sales.
- Recommendation: do not change the rule on this evidence. Run a controlled trial on the highest-waste items, and start recording order quantity, noon sales, sell-out time and defrost quantity so that lost sales can be measured properly (Section 8).

## 2. Data and method

Data: 43 items, 92 days, 3,894 daily rows (open, receive, sold, mark out, closing, variance, mark-out verified) loaded into MySQL. Fresh items arrive every morning. Frozen items (cakes, cake pops, Bearista bears, triple chocolate cookie, egg bites) arrive on Fridays and are defrosted in store.

Definitions agreed for this analysis:

- Mark out: stock thrown away because it expired, valued at cost price. Verified and unverified mark outs both count as waste. Variance (a unit missing from the physical count) is reported separately and is never added to mark out.
- Waste rate, primary: mark out cost divided by the cost of stock handled (opening + received). Because opening stock is counted again each day it is held, this is a stock-days measure and it makes long-held frozen stock look lower than it is.
- Waste rate, secondary: mark out cost divided by the cost of sales. A third measure, mark out as a share of stock received, is kept as a rough reference only.
- Stock-out day, items with a shelf life of 2 days or more: closing is 0 and mark out is 0. Closing 0 with a mark out means the stock expired, so it is not a stock-out.
- Stock-out day, the 1-day item (I013 Dark Chocolate Donut): opening plus received is above 0 and mark out is 0.
- Stock-outs are counted in days, never in units. A closing of 0 can also mean the last unit sold at closing, so each one is a possible lost-sales day, except the 19 days with no stock at all.

## 3. Data quality

Checks were run with SQL on the loaded tables. Nothing was changed; the issues below are flagged and the rows are kept as recorded.

| Check | Result |
|---|---|
| Row counts, duplicates, orphan items and dates, negative quantities | All as expected: 3,894 rows, no duplicates, none orphaned, none negative |
| Sheet balance rule (open + receive - sold - mark out + variance = closing) | 3 rows fail (table below) |
| Opening stock vs previous closing | 2 rows differ, both knock-on effects of the first two balance errors |
| Fresh-item rules | No exceptions: I013 opens and closes at 0 every day; 2-day items mark out exactly what is left from yesterday |
| Mark outs not ticked the next morning | 18 rows (44 units, 305.51) with mo_verified = 0; kept in waste |
| Variance | Only 2 rows, both -1 unit (7 Sep I007, 29 Sep I020); -16.60 at cost; excluded from waste |
| Outliers in sold, receive, mark out | Nothing conclusive; no obvious typing errors, but the check is weak on small counts |

Rows that do not balance:

| Date | Item | Closing on sheet | Closing by the sheet rule |
|---|---|---|---|
| 19 Jul 2026 | I012 | 36 | 38 |
| 29 Sep 2026 | I001 | 22 | 19 |
| 30 Sep 2026 | I012 | 6 | 9 |

For the first two rows, the next day's opening matches the rule's number, which suggests a mistyped closing. This could not be confirmed, and the third row has no next day to compare. None of the three changes a stock-out flag, because closing is above 0 either way. Mark out is affected only if the mark out cell itself was the wrong one; both rows record 0.

Limits of the data:

- Edges: opening stock on 1 July comes from June, and stock left on 30 September (31 of 43 items) will be marked out in October. Waste is slightly understated at both ends. The first and last weeks are partial and are included.
- Frozen items: the log has no record of units or dates defrosted, and closing includes freezer stock. An item can be empty on display while the freezer still holds units, so frozen stock-outs are undercounted. Five frozen items show zero stock-out days.
- Item names such as "Dainsh" and "Siracha" were left as they appear in the source. Chicken Siracha Flatbread (I029) exists from 1 September only.
- No order quantity or hourly sales are recorded. For fresh items, the order for day D is taken to be the quantity received on day D+1.

## 4. Waste

Total mark out: 21,435.83 at cost, 3,126 units. Fresh items are 17,535.88 (81.8%) and frozen items 3,899.95 (18.2%). The 44 unverified units (305.51) are included.

| Category | Cost | % of total | % of stock handled | % of sales |
|---|---|---|---|---|
| Sandwich | 6,612.50 | 30.8 | 12.8 | 25.2 |
| Pastry | 4,824.74 | 22.5 | 8.7 | 15.1 |
| Cakes | 3,234.34 | 15.1 | 2.4 | 24.8 |
| Salad | 2,805.16 | 13.1 | 26.7 | 94.3 |
| Cookies | 1,436.48 | 6.7 | 8.7 | 21.1 |
| Donuts | 1,169.50 | 5.5 | 15.5 | 26.9 |
| Muffins | 777.00 | 3.6 | 22.7 | 66.1 |
| Savory | 321.11 | 1.5 | 0.9 | 9.2 |
| Juice | 255.00 | 1.2 | 8.3 | 14.3 |
| All items | 21,435.83 | 100.0 | 6.7 | 23.3 |

Ten items account for about 40% of waste. Cost and rate tell different stories: Cheese Croissant has many units but a low rate, while Dark Chocolate Donut has the most units of any item (201) at 47.0% of what it handled.

| Item | Units | Cost | Units % of received |
|---|---|---|---|
| Halloumi Flat Bread | 121 | 1,210.00 | 12.8 |
| Grand Chicken Caesar Salad | 89 | 1,076.90 | 52.0 |
| Chicken Pesto Salad | 76 | 1,026.00 | 47.2 |
| Turkey & Cheese Pretzel Supreme | 128 | 947.20 | 12.9 |
| Chicken Caesar Wrap | 71 | 781.00 | 29.7 |
| Tuna Flat Bread | 86 | 731.00 | 50.0 |
| Chicken Avocado Protein Wrap | 65 | 728.00 | 26.7 |
| Oreo and Chocolate Cake | 56 | 719.60 | 30.4 |
| Cheese Croissant | 142 | 714.26 | 8.1 |
| Cheese and Roasted Tomato | 82 | 705.20 | 16.8 |

| Period | Days | Cost | Cost per day | % of stock handled |
|---|---|---|---|---|
| Mon-Thu | 53 | 15,536.97 | 293.15 | 9.5 |
| Fri-Sun | 39 | 5,898.86 | 151.25 | 3.7 |
| July | 31 | 6,626.08 | 213.74 | 5.9 |
| August | 31 | 7,452.26 | 240.40 | 6.5 |
| September | 30 | 7,357.49 | 245.25 | 7.9 |

By day, Tuesday is highest at 394.04 per day and Monday next at 346.29; Friday is lowest at 145.69. The waste rate rises each month even though September sales were quieter.

## 5. Stock-outs

Total stock-out days: 671 of 3,894 item-days (17.2%).

- No stock all day (opening 0 and received 0): 19 days, all frozen items. These are certain lost-sales days.
- Closing-zero possible stock-out: 652 days.

- By supply type: fresh items 638 of 2,698 (23.6%), including I013 at 25 of 92 days (27.2%); frozen items 33 of 1,196 (2.8%). The frozen figure is undercounted for the reason given in Section 3.

| Category | Stock-out days | Share of item-days |
|---|---|---|
| Donuts | 49 | 26.6% |
| Pastry | 244 | 26.5% |
| Juice | 24 | 26.1% |
| Sandwich | 200 | 23.3% |
| Cookies | 58 | 21.0% |
| Muffins | 32 | 17.4% |
| Salad | 37 | 13.4% |
| Savory | 5 | 2.7% |
| Cakes | 22 | 2.4% |

| Weekday | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| Stock-out share of item-days | 7.5% | 12.5% | 16.5% | 21.5% | 24.9% | 20.4% | 17.5% |

Fri-Sun is 20.9% (345 of 1,650 item-days) against 14.5% for Mon-Thu (326 of 2,244). By month: July 17.7%, August 16.7%, September 17.2%. The most affected items are Pain au Chocolate (30 of 92 days), Turkey Cheese Sandwich (29) and Plain Croissant (27). Waste is highest on Monday and Tuesday while stock-outs are highest on Thursday and Friday.

## 6. Ordering rule test

Rule as described by the store: an order placed by noon on day D arrives on D+1 and equals sales of the previous day (D-1) plus 10% if D is Monday to Thursday, or plus 30% if D is Friday to Sunday. If the previous day had no sales, the order stays the same as the previous order. Tested on the 30 fresh items, 2,638 orders, rounded to the nearest unit.

| Measure | Result |
|---|---|
| Units ordered, actual vs rule | 13,873 vs 13,986 (-0.8%) |
| Exact match to the rule | 22.4% |
| Within 1 unit of the rule | 39.3% |
| Above / below the rule | 36.5% / 33.6% |
| Typical gap | 2 units (median), 3.25 (average) |
| Orders after a zero-sales day equal to the previous order | 169 of 297 (56.9%) |

| Order placed on | Rule | Actual (median multiple of D-1 sales) | Actual vs rule, units |
|---|---|---|---|
| Monday | 1.10 | 0.88 | 1,563 vs 2,373 (-34%) |
| Tuesday | 1.10 | 1.00 | 1,514 vs 1,608 |
| Wednesday | 1.10 | 1.00 | 1,481 vs 1,557 |
| Thursday | 1.10 | 1.18 | 1,947 vs 1,577 (+23%) |
| Friday | 1.30 | 1.60 | 2,444 vs 1,820 (+34%) |
| Saturday | 1.30 | 1.33 | 2,613 vs 2,457 |
| Sunday | 1.30 | 1.00 | 2,311 vs 2,595 (-11%) |

Result: the 10% / 30% rule cannot be confirmed from the log alone. Total units are close, but the day-by-day pattern differs, and the log lacks the noon sales needed to rebuild the orders. Using same-day sales instead of D-1 fits only slightly better (23.5% exact), so the basis the store really uses cannot be identified from this data. Management should also note that the store sometimes does not raise orders, or lowers them, when sales are low or zero.

## 7. Alternative rules: stock replay

To see what other ordering rules would have done, stock was replayed day by day for each item under each rule, with recorded sales as the demand. Scope: 29 fresh items (I029 excluded for lack of history), 29 July to 30 September, 1,856 item-days. Replaying the actual orders reproduces recorded waste to within 7.40, the one variance unit on I007. "Short days" are item-days where recorded sales were higher than the stock the rule would have had; they are certain missed-sales days, and are 0 for the actual orders by construction.

| Rule | Waste | Possible stock-out days | Short days |
|---|---|---|---|
| Actual orders | 12,250.90 | 431 | 0 |
| Plain 10% / 30% on D-1 sales | 22,467.70 | 124 | 69 |
| Same weekday, last 2 weeks average | 9,251.00 | 473 | 264 |
| Same weekday, last 4 weeks average | 10,244.84 | 349 | 167 |
| Same weekday, last 4 weeks, +10% | 15,923.43 | 183 | 84 |

- If the store followed the plain rule on D-1 sales, waste would be 83% higher than recorded, with far fewer stock-outs.
- The same-weekday rule saves most on Tuesday (waste 1,439 vs 3,012) and Wednesday (1,703 vs 2,151), and is worse on Monday (2,397 vs 2,250).
- Because the rule rounds up to a whole unit, even a small buffer adds a unit on many items (4-week rule: waste 10,245 at +0%, 12,900 at +2%).

How far to trust this: recorded sales are capped by the stock the store actually had, so they understate demand on stock-out days. A rule that orders more than the store did cannot show extra sales, and its waste may be overstated; a rule that orders less shows missed sales correctly. The table is therefore a trade-off, not a ranking. The rules were not fitted to the data, but only a small set was tried.

## 8. Recommendations

These follow from the evidence above and are limited by it. They are hypotheses to be tested in the store, not conclusions.

- Do not change the ordering rule on the strength of this analysis. No simple rule reduced waste without also missing recorded sales.

- Run a controlled trial for about four weeks on the highest-waste items (for example Halloumi Flat Bread, Grand Chicken Caesar Salad and Chicken Pesto Salad). For orders placed Monday to Wednesday, base the order on the same weekday of recent weeks instead of yesterday's sales, and compare waste and sold-out days against the current method.
- Review salads and muffins first. Salad waste is 26.7% of stock handled and nearly equals the cost of what is sold (94.3%). Check order sizes against sales for these items.
- Review the Thursday order for Friday arrival. Friday has the highest stock-out share (24.9%), while Thursday orders already ran 23% above the plain rule.
- Start recording order quantity, sales up to noon, the time an item sells out and the units and date defrosted. Without these, lost sales cannot be measured in units and the ordering rule cannot be tested directly.
- Tighten the daily sheet: check the balance rule at close, tick mark outs the next morning, and investigate every variance.

## 9. Limitations

- Results cover one store over one quarter (92 days), so seasonal patterns beyond this period cannot be assessed.
- Stock-out days are possible lost sales, not measured lost sales, and are counted in days only.
- Frozen stock cannot be tracked accurately without defrost records.
- Edge effects and partial weeks affect waste slightly at the start and end of the quarter.
- The ordering and replay results depend on recorded sales as a demand proxy.

## Appendix: How to verify the main figures in MySQL

| Figure | Query to run | Expected |
|---|---|---|
| Total mark out | SELECT SUM(mark_out_qty), ROUND(SUM(mark_out_cost),2) FROM v_food_log; | 3,126 units, 21,435.83 |
| Unbalanced rows | SELECT * FROM v_balance_check; | 3 rows |
| Unticked mark outs | SELECT * FROM daily_food_log WHERE mo_verified = 0; | 18 rows |
| Stock-out days | v_stockout view; SUM(stockout_day) over all rows | 671 of 3,894 |
| Ordering test | v_order_test view; compare actual_order with rule_order | 2,638 rows |
| Stock replay | step6_replay.sql | Actual 12,250.90 / 431 / 0; 4-week rule 10,244.84 / 349 / 167 |